

EDA

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```
library(tidyverse)
library(ggplot2)
library(dplyr)
library(corrplot)
```

Data Summary

Our data is from the college scorecard data, [here](#)

We are looking at the most recent field of study dataset. This consists of 229188 observations with 174 columns, regarding students in the 2019/2020 school year.

Our institutional dataset is also pretty big, with 6694 rows and 3306 columns. We will join these dataframes on the university id, after reducing them in size.

This is how we loaded our data in.

```
data <- read_csv("Most-Recent-Cohorts-Field-of-Study.csv") #on individuals
institution_data <- read_csv("MERGED2019_20_PP.csv") #on institutions
```

We then joined using left join.

```
joined_df <- institution_data %>%
  left_join(data, by = "OPEID6")
```

This gave us a monstrous dataframe, with 3500 columns, and over 200000 rows. To reduce our data to a more manageable size, we did the following.

```

df_clean <- joined_df[, colMeans(is.na(joined_df)) <= 0.50]
df_clean <- df_clean %>%
  select(where(~ n_distinct(.) > 1))

threshold <- 0.01

df <- df_clean %>%
  select(where(~ {
    if (!is.numeric(.x)) {
      TRUE
    } else {
      v <- var(.x, na.rm = TRUE)
      !is.na(v) && v >= threshold
    }
  })))

```

This still had over 2000 columns. So what we did was we chose X many predictors that we thought could be interesting, to investigate further.

```

interesting_col_names <- c(
  "EARN_MDN_4YR",
  "OPEID6",
  "CITY",
  "SCH_DEG",
  "MAIN.x",
  "NUMBRANCH",
  "PREDEG",
  "HIGHDEG",
  "ST_FIPS",
  "REGION",
  "ADM_RATE_ALL",
  "PCIP12",
  "PCIP24",
  "PCIP51",
  "PCIP52",
  "UGDS",
  "UG25ABV",
  "UGDS_WHITE",
  "UGDS_BLACK",
  "UGDS_HISP",
  "PPTUG_EF",
  "COSTT4_A",

```

```

"TUITIONFEE_IN",
"TUITIONFEE_OUT",
"TUITFTE",
"INEXPTE",
"AVGFACSA",
"PFTFAC",
"C150_4",
"PFTFTUG1_EF",
"D150_4",
"C150_4_WHITE",
"C150_4_BLACK",
"C150_4_HISP",
"DEATH_YR8_RT",
"COMP_ORIG_YR8_RT",
"COMP_4YR_TRANS_YR8_RT",
"PELL_COMP_4YR_TRANS_YR8_RT",
"PELL_WDRAW_ORIG_YR8_RT",
"NOPELL_WDRAW_ORIG_YR8_RT",
"NOT1STGEN_COMP_4YR_TRANS_YR2_RT",
"FIRSTGEN_COMP_ORIG_YR2_RT",
"TUITFTE",
"OVERALL_YR2_N",
"DEBT_MDN",
"CREDESC",
"DEBT_ALL_STGP_ANY_MEAN",
"BBRR1_FED_UG_DFLT",
"DISTANCE",
"CONTROL.x",
"CONTROL.y",
"SATVRMID",
"SATMTMID"
)

```

We then created a dataframe with those interesting attributes. However, we were think about using career earnings as an outcome, thus we only are interested on results with a numeric earnings. Thus we get rid of all rows that have a character or na value in `EARN_MDN_4YR` column, or career earnings after 4 years.

But we still had the issue that there were many different columns with many NA values, some close to 40%. For simplicity, we decided to drop NA values. This has the potential to introduce bias in our data, because there could be some other factor influencing career earnings and whether or not there are many NA values.

```
very_trimmed_df <- df[interesting_col_names]
```

```
final_df <- very_trimmed_df[!is.na(as.numeric(as.character(very_trimmed_df$EARN_MDN_4YR))), ]
```

```
na_df <- na.omit(final_df)
```

All of this was done on a different document. Loading in the result from that process gives us the following dataset.

```
df <- read.csv("final_data.csv")
```

We still have some decisions to make regarding what data to use. Much of the variables we have are 75% or more just PS, which likely means primarily suppressed, which means not included since the sample size was too small, or the data was private. Since many of the predictors consist of things like debt after college, it makes sense that it is disclosed to the public (people like us). Our final result is a dataframe consisting of 52 columns, and roughly 30000 observations, which is a much more manageable size.

EDA Begins

The first thing we will do in our EDA is plot a correlation matrix. This gives us an idea of what predictors are similar to others, and which are uncorrelated. The key points to remember are that our outcome variable is EARN_MDN_4YR, so predictors with stronger correlation to it are likely good predictors.

```
df %>% select(is.numeric, - OPEID6) %>% #only continuous output, without passenger id
cor(use = "pairwise.complete.obs") %>%
corrplot(type='lower',
          method = "color",          # Use colored squares/cells for compactness
          order = "hclust",          # Cluster similar variables together
          tl.cex = 0.5,               # Shrink text label size (adjust as needed)
          tl.col = "black",           # Text label color
          tl.srt = 45,               # Rotate text labels 45 degrees
          diag = FALSE)              # Hide self-correlations on the diagonal
```

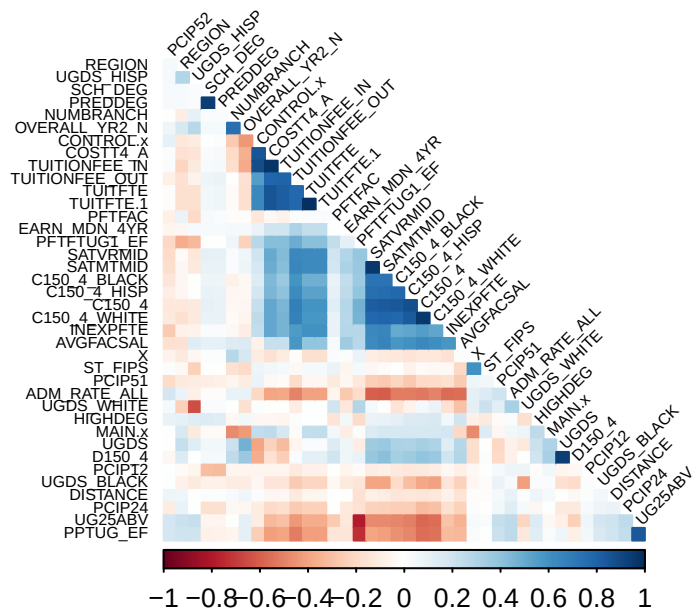
Warning: Use of bare predicate functions was deprecated in tidysselect 1.1.0.

i Please use wrap predicates in `where()` instead.

Was:

```
data %>% select(is.numeric)
```

```
# Now:
data %>% select(where(is.numeric))
```



Choosing better predictors

For a quick way to get rid of some strongly correlated predictors, we fit an AIC model, and saw which variables were least important.

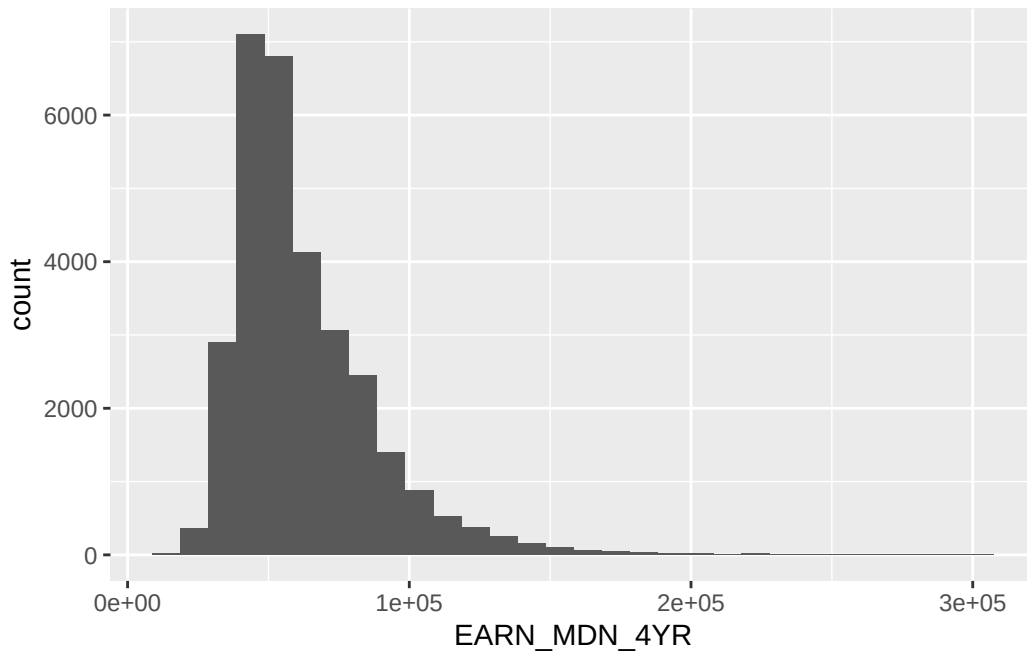
Some plots

Now lets quickly investigate a few variables relationship with the outcome variable.

First, lets look at the distribution of the outcome variable. We see that it looks to be roughly lognormally distributed. This is reasonable for a outcome variable.

```
ggplot(df, aes(x = EARN_MDN_4YR)) +
  geom_histogram()
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



Admission Rate versus Earnings

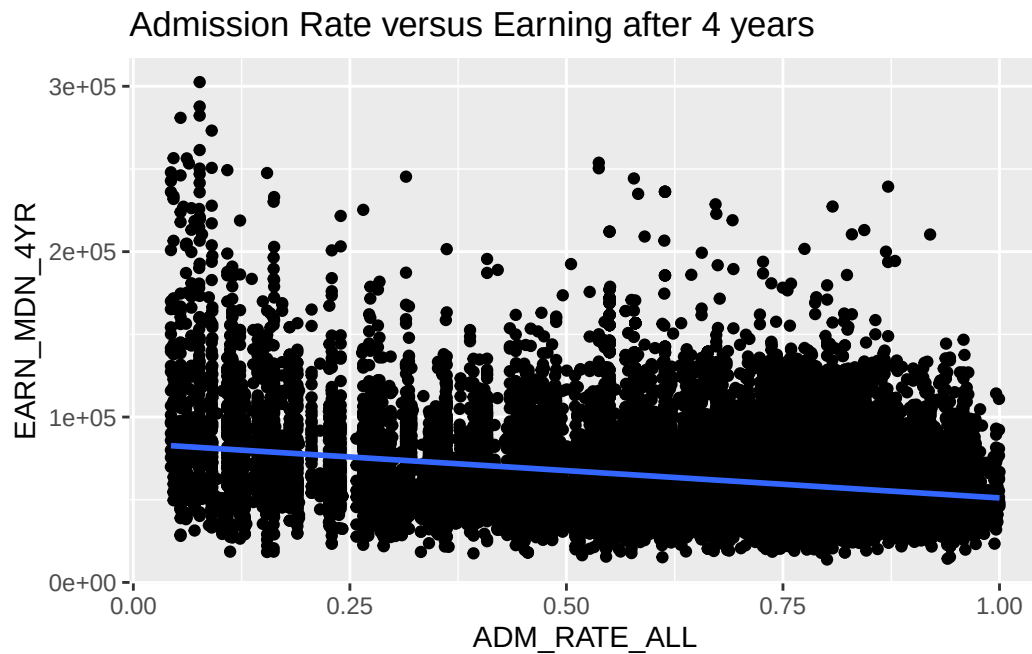
One thing college hopefuls are aware of when applying to college is admission rate, where it is generally thought that better schools have lower admission rates. Below is the scatterplot. We can see that there seems to be a meaningful higher wages for very low acceptance rates, under .15. It also looks like majority of the data has an acceptance rate between .5 and .9. The correlation between these two variables is $-.26$, which suggests this is a reasonable predictor.

Looking at the fitted simple linear model summary, we have an R squared value of 0.06823, which is not great, given how many predictors we have, is fairly good. The p values are very small, but that just means it is unlikely the data comes from a constant model. Given our sample size is so big, even small variations from constant lead to tiny p values.

```
simple_df <- df[c("EARN_MDN_4YR", "ADM_RATE_ALL")]
simple_model <- lm(EARN_MDN_4YR~ADM_RATE_ALL, simple_df)

ggplot(df, aes(x=ADM_RATE_ALL, y = EARN_MDN_4YR)) +
  geom_point()+
  labs(
    title = "Admission Rate versus Earning after 4 years"
  ) +
  geom_smooth(method = "lm")
```

```
`geom_smooth()` using formula = 'y ~ x'
```



```
cor(df$EARN_MDN_4YR, df$ADM_RATE_ALL)
```

```
[1] -0.2612132
```

```
summary(simple_model)
```

Call:

```
lm(formula = EARN_MDN_4YR ~ ADM_RATE_ALL, data = simple_df)
```

Residuals:

Min	1Q	Median	3Q	Max
-61827	-16088	-6036	11571	220935

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	84121.7	479.8	175.34	<2e-16 ***
ADM_RATE_ALL	-33036.3	696.3	-47.45	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 24590 on 30739 degrees of freedom
Multiple R-squared: 0.06823, Adjusted R-squared: 0.0682
F-statistic: 2251 on 1 and 30739 DF, p-value: < 2.2e-16

Debt versus earnings

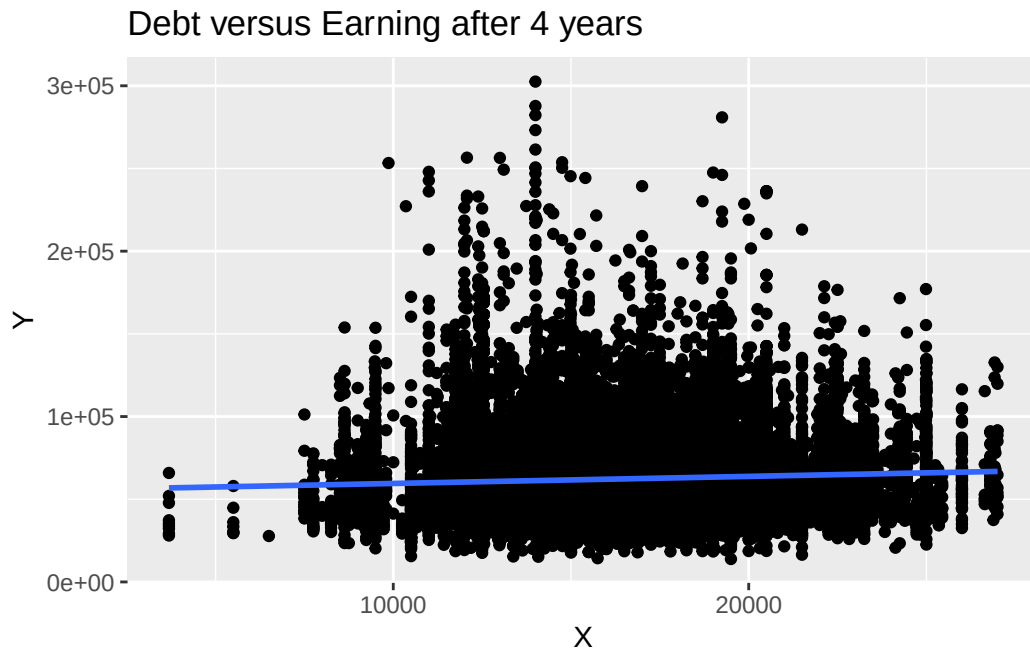
One predictor we would like to use is debt. However, not everyone has a recorded debt. So for eda, we will filter for having a numeric value, and impute later. The given slope is small. The R squared value for the simple linear model is 0.003076, which means the model accounts for 0.3 percent of variation from the mean, which isn't great.

```
debt_df <- tibble(  
  Y=df$EARN_MDN_4YR,  
  X=as.numeric(df$DEBT_MDN)  
)
```

Warning in eval_tidy(xs[[j]], mask): NAs introduced by coercion

```
debt_df <- debt_df %>% filter(X != "PS")  
ggplot(debt_df, aes(x=X, y = Y)) +  
  geom_point()+  
  labs(  
    title = "Debt versus Earning after 4 years"  
  ) +  
  geom_smooth(method="lm")
```

`geom\_smooth()` using formula = 'y ~ x'



```
debt_model <- lm(Y~X, debt_df)
summary(debt_model)
```

Call:

```
lm(formula = Y ~ X, data = debt_df)
```

Residuals:

Min	1Q	Median	3Q	Max
-49758	-17048	-6840	11734	241270

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	5.527e+04	7.425e+02	74.438	<2e-16 ***
X	4.276e-01	4.391e-02	9.738	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 25440 on 30731 degrees of freedom

Multiple R-squared: 0.003076, Adjusted R-squared: 0.003044

F-statistic: 94.82 on 1 and 30731 DF, p-value: < 2.2e-16

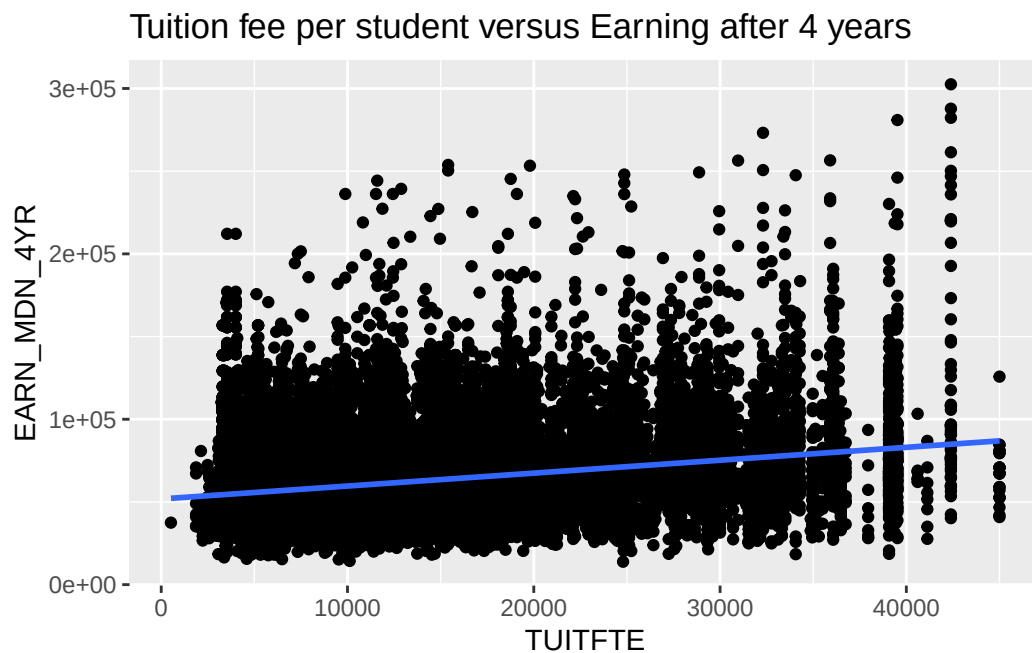
```
#cor(df$DEBT_MDN, df$EARN_MDN_4YR)
```

Net Tuition Revenue per student versus Earnings

One thing that feels like it should correlate with good schools is tuition, where better schools cost more. Looking at the fitted linear model, we see that it has an R squared of 0.06083, which means it accounts for about 6 percent of error from the mean is explained by net tuition.

```
ggplot(df, aes(x=TUITFTE, y = EARN_MDN_4YR)) +  
  geom_point()+  
  labs(  
    title = "Tuition fee per student versus Earning after 4 years"  
  ) +  
  geom_smooth(method="lm")
```

`geom\_smooth()` using formula = 'y ~ x'



```
tuit_model <- lm(EARN_MDN_4YR ~ TUITFTE, df)  
summary(tuit_model)
```

Call:

```
lm(formula = EARN_MDN_4YR ~ TUITFTE, data = df)
```

Residuals:

Min	1Q	Median	3Q	Max
-63692	-16487	-6222	11296	217630

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	5.181e+04	2.751e+02	188.31	<2e-16 ***
TUITFTE	7.805e-01	1.749e-02	44.62	<2e-16 ***

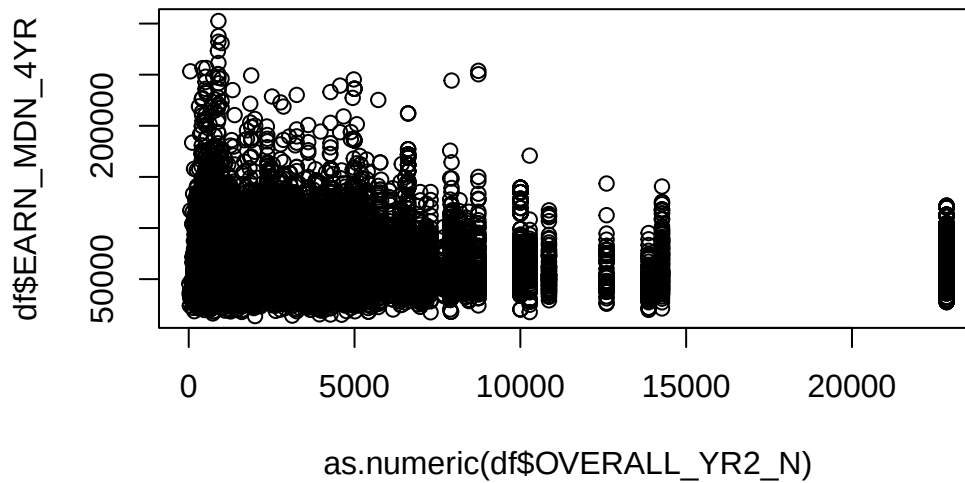
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 24690 on 30739 degrees of freedom

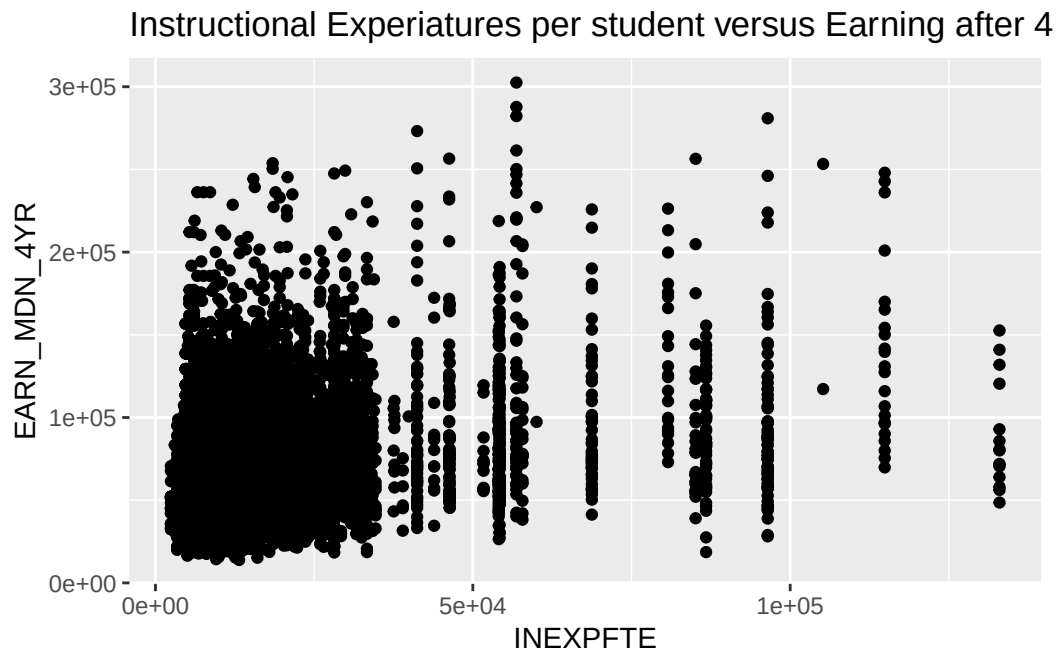
Multiple R-squared: 0.06083, Adjusted R-squared: 0.0608

F-statistic: 1991 on 1 and 30739 DF, p-value: < 2.2e-16

```
plot(as.numeric(df$OVERALL_YR2_N), df$EARN_MDN_4YR)
```

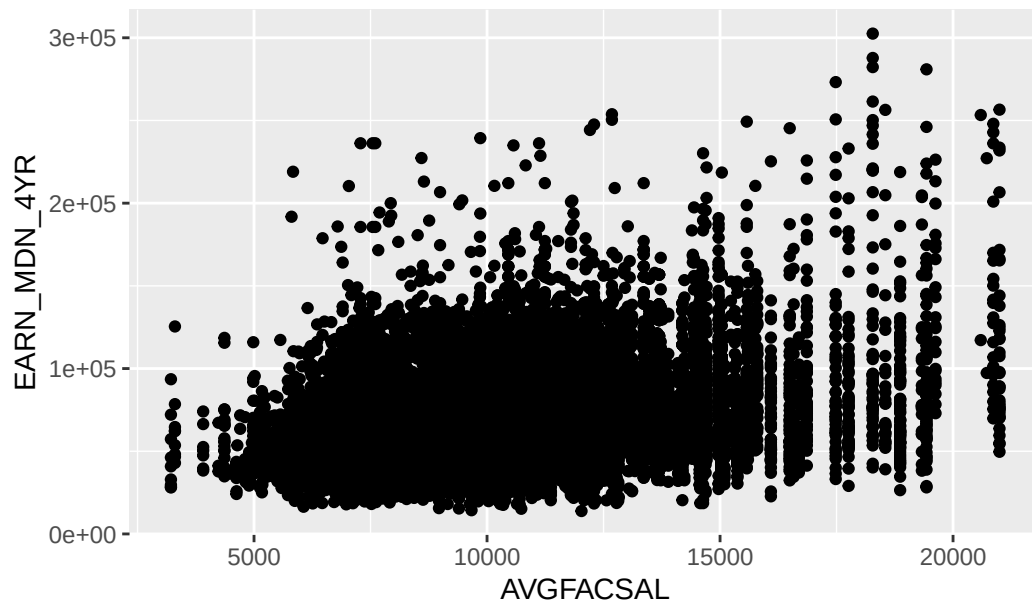


```
ggplot(df, aes(x=INEXPFTE, y = EARN_MDN_4YR)) +
  geom_point()+
  labs(
    title = "Instructional Experiatures per student versus Earning after 4 years"
  )
```

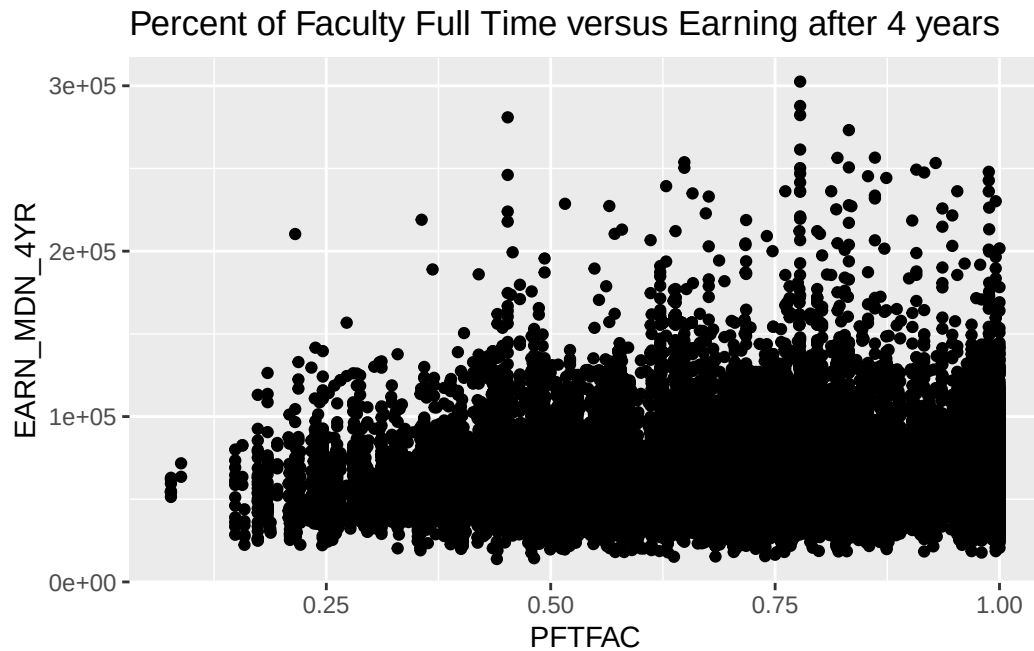


```
ggplot(df, aes(x=AVGFACSAL, y = EARN_MDN_4YR)) +
  geom_point()+
  labs(
    title = "Average Faculty Salary versus Earning after 4 years"
  )
```

Average Faculty Salary versus Earning after 4 years



```
ggplot(df, aes(x=PFTFAC, y = EARN_MDN_4YR)) +  
  geom_point()+  
  labs(  
    title = "Percent of Faculty Full Time versus Earning after 4 years"  
  )
```



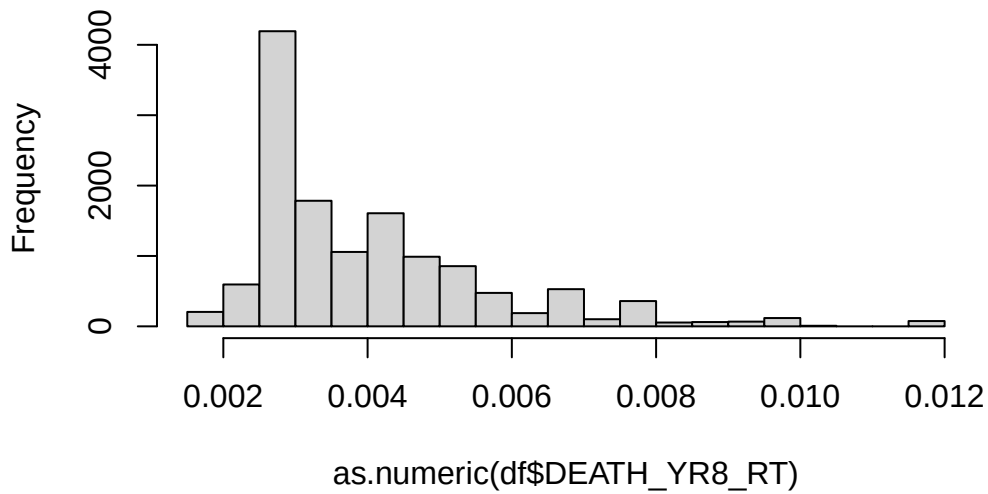
Death Rate

One predictor we were interested in was death rate. One issue we ran into was much of the data was recorded as PS, instead of a numeric value. For now, let's just look at the datapoints with a non PS value. The histogram of outputs looks to be roughly lognormal (or some other right skewed distribution). Plotting out these points, we can see a real relationship.

```
hist(as.numeric(df$DEATH_YR8_RT)) #need to convert
```

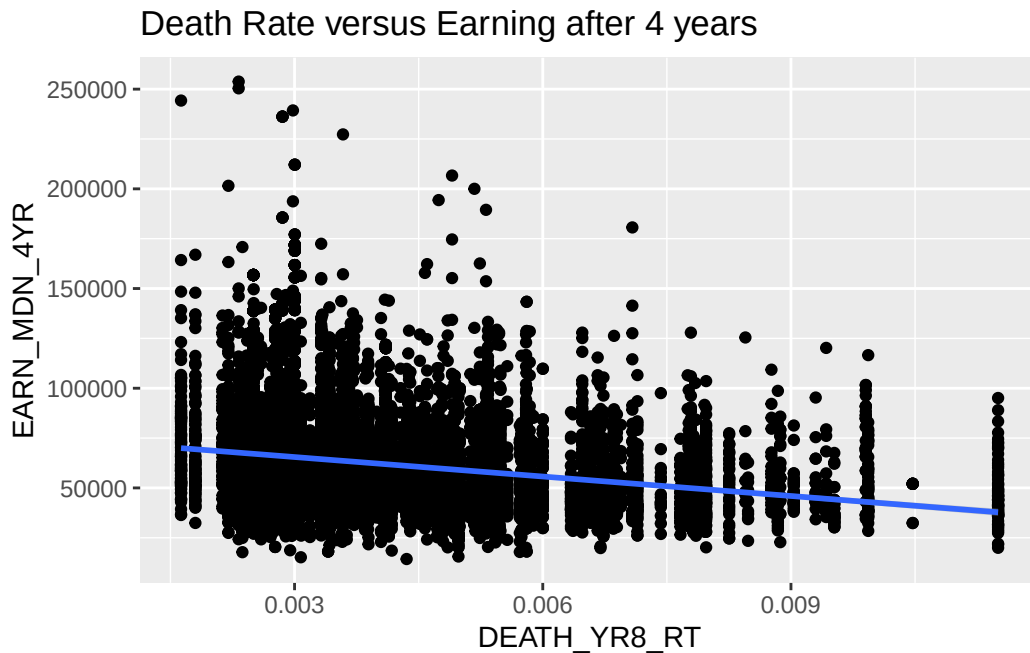
Warning in `hist(as.numeric(df$DEATH_YR8_RT))`: NAs introduced by coercion

Histogram of as.numeric(df\$DEATH_YR8_RT)



```
dead_df <- df %>% filter(DEATH_YR8_RT != "PS")
dead_df$DEATH_YR8_RT <- as.numeric(dead_df$DEATH_YR8_RT)
ggplot(dead_df, aes(x=DEATH_YR8_RT, y = EARN_MDN_4YR)) +
  geom_point()+
  labs(
    title = "Death Rate versus Earning after 4 years"
  ) +
  geom_smooth(method="lm")
```

``geom_smooth()`` using formula = `'y ~ x'`



```
cor(dead_df$DEATH_YR8_RT, dead_df$EARN_MDN_4YR)
```

```
[1] -0.2329507
```

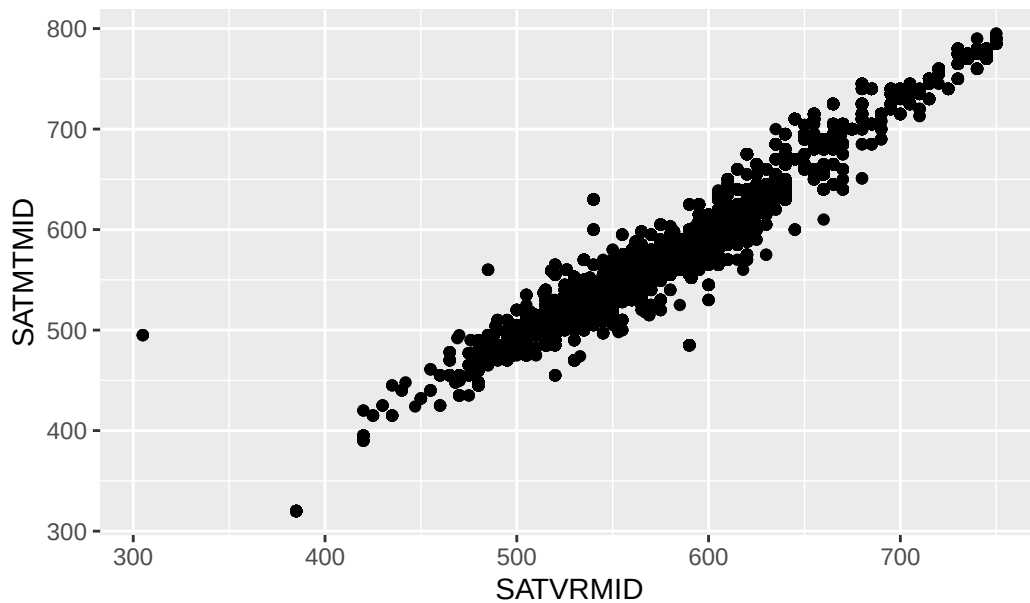
SAT SCORES

We had two different SAT scores, median critical reading, and median math scores for the college. These have a very strong correlation, so we probably cannot use both as they are.

One potential way we could use both is using a linear combination of both. We did to linear combinations, $X1 = M + R$ and $X2 = M - R$, which look to be mostly uncorrelated. We also fit a linear model using this in the appendix, which has an R squared value of 0.09969, which considering we only used 2 highly correlated predictors from 50, that fairly good.

```
ggplot(df, aes(x=SATVRMID, y = SATMTMID)) +
  geom_point()+
  labs(
    title = "SAT Median Critical Reading Score versus SAT Median Math Score"
  )
```

SAT Median Critical Reading Score versus SAT Median Math Score

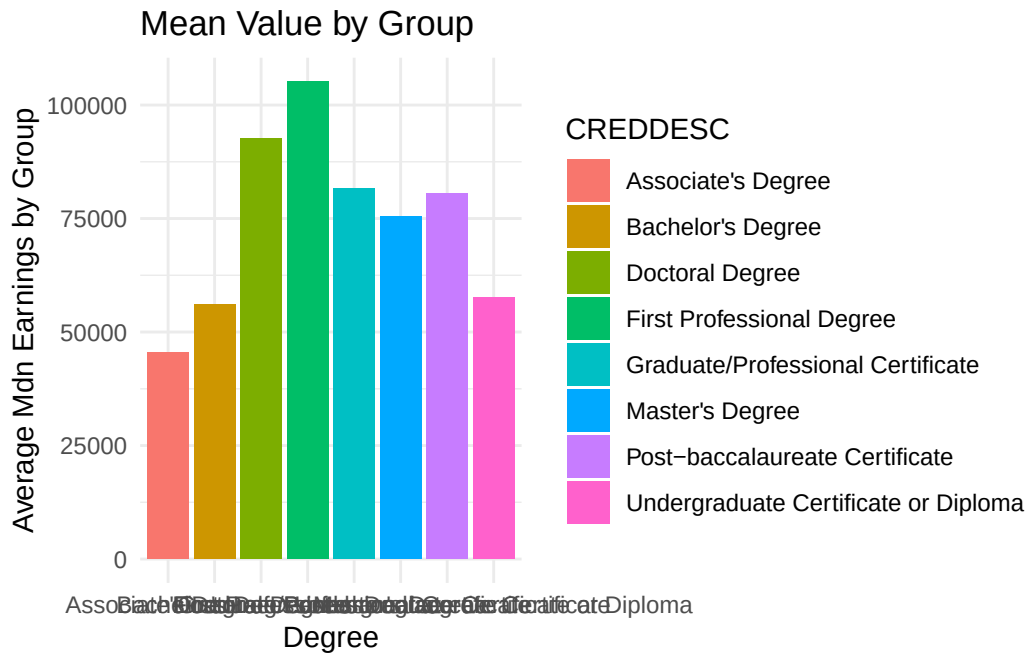


Earnings by Educational Level

We can look at the earning levels by educational level. We can see that more education tends to lead to more earnings, as expected

```
new_df <- df
new_df %>%
  # 1. Group the data and calculate the mean for each group
  group_by(CREDESC) %>%
  summarize(mean_value = mean(EARN_MDN_4YR, na.rm = TRUE)) %>%

  # 2. Pass the aggregated data directly into ggplot
  ggplot(aes(x = CREDESC, y = mean_value, fill = CREDESC)) +
  geom_col() + # geom_col() plots the actual values provided in y
  labs(
    title = "Mean Value by Group",
    x = "Degree",
    y = "Average Mdn Earnings by Group"
  ) +
  theme_minimal()
```



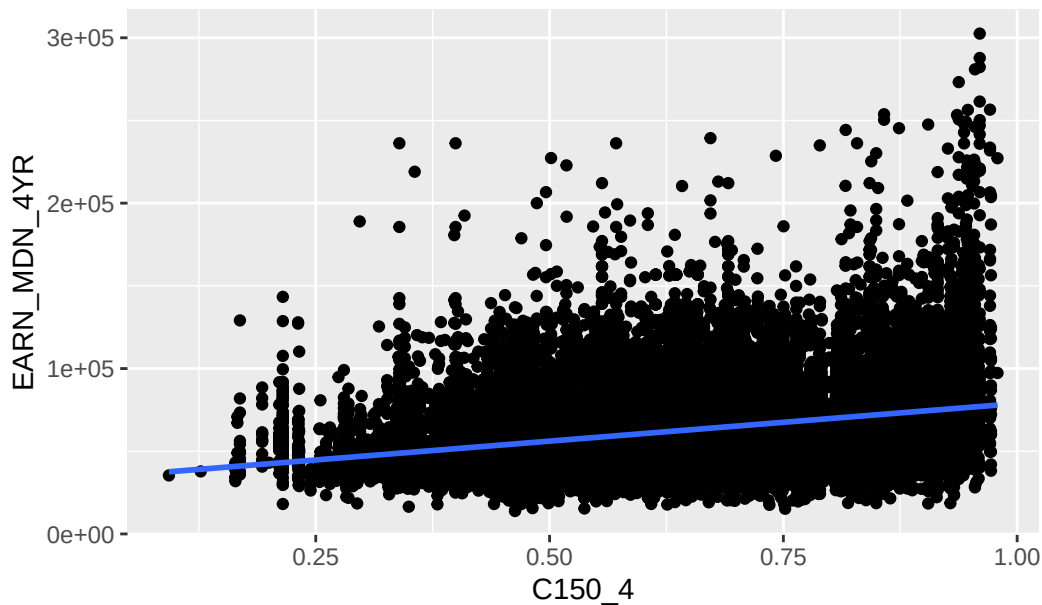
First Time Student Completion Rate

Looking through the data, we were curious at what first time student completion rate meant in terms of career earning. We also saw different variables for different demographics. Fitting a simple linear model, we get an r squared value of 0.07988169.

```
ggplot(df, aes(x=C150_4 , y = EARN_MDN_4YR)) +
  geom_point()+
  labs(
    title = "First Time Student Completion Rate vs Earnings"
  ) +
  geom_smooth(method="lm")
```

`geom\_smooth()` using formula = 'y ~ x'

First Time Student Completion Rate vs Earnings



```
hmmm <- lm(EARN_MDN_4YR~C150_4, df)
summary(hmmm)$r.squared
```

```
[1] 0.07988169
```

TODO

Convert the following to numeric

One thing we ran into when using the data from individuals is that many of them have private information. This was recorded as PS, rather than a number recorded. So for many predictors, such as debt, we have to decide how to deal with them. One way is to impute the data. We could impute using mean imputation, where we set every PS values as the mean, either mean of everyone or mean of group, or we could predict what they are based off of other values, probably using a prebuilt function. We could also set them all to zero. We could also drop all the PS values.

Ruling out drop PS

When we look at the data, some of the predictors with PS can contain up to 90 percent of datapoints containing PS. If we through all of those points out, we would lose most of our

data. Thus dropping all observations with PS would work for at most 1 variable, with only 50 of observations recorded as PS.

Selecting few of these

Due to the difficulties in working with much of this data, we are only going to work with 2-4 of these predictors.

```
DEATH_YR8_RT, COMP_ORIG_YR8_RT, COMP_4YR_TRANS_YR8_RT, PELL_COMP_4YR_TRA  
PELL_WDRAW_ORIG_YR8_RT, NOPELL_WDRAW_ORIG_YR8_RT, NOT1STGEN_COMP_4YR_T  
FIRSTGEN_COMP_ORIG_YR2_RT, OVERALL_YR2_N, DEBT_MDN, DEBT_ALL_STGP_ANY_ME  
BBRR1_FED_UG_DFLT
```

```
#str(df)
```

Multicollinearity and Related Simple Linear Models

There are several items which have especially high correlation. As such, we are generally concerned with removing them to prevent any multicollinearity. Among these is are C150_4 and C150_4_WHITE, which have high correlation. For these, the correlation and fitted simple linear model are as follows:

```
cor(df$C150_4, df$C150_4_WHITE)
```

```
[1] 0.9653627
```

```
summary(lm(C150_4 ~ C150_4_WHITE, data = df))
```

Call:

```
lm(formula = C150_4 ~ C150_4_WHITE, data = df)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.49757	-0.01265	0.00425	0.01475	0.52210

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.009100	0.001024	-8.89	<2e-16 ***
C150_4_WHITE	0.980974	0.001512	648.70	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.04126 on 30739 degrees of freedom
Multiple R-squared: 0.9319, Adjusted R-squared: 0.9319
F-statistic: 4.208e+05 on 1 and 30739 DF, p-value: < 2.2e-16

This makes sense, as the total C150_4 value is predominantly from white students, and minorities are related to this by direct evaluation.

On the other hand COSTT4_A and TUITIONFEE_IN are also highly correlated. This is because cost of attendance is highly correlated to in-state tuition.

```
cor(df$COSTT4_A, df$TUITIONFEE_IN)
```

```
[1] 0.9783546
```

```
summary(lm(TUITIONFEE_IN ~ COSTT4_A, data = df))
```

Call:

```
lm(formula = TUITIONFEE_IN ~ COSTT4_A, data = df)
```

Residuals:

Min	1Q	Median	3Q	Max
-17073.5	-2012.4	-239.2	1539.7	9380.7

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.058e+04	4.100e+01	-258.0	<2e-16 ***
COSTT4_A	9.207e-01	1.111e-03	828.9	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3149 on 30739 degrees of freedom
Multiple R-squared: 0.9572, Adjusted R-squared: 0.9572
F-statistic: 6.871e+05 on 1 and 30739 DF, p-value: < 2.2e-16

Finally, PFTFTUG1_EF and UG25ABV are also highly correlated. This is because one is the percent of undergraduate that are working part time, and the other is the percent of undergraduates under 25 years old. Given most people who do undergraduate education past 25 have to support themselves, this seems reasonable. Here, we omit PFTFTUG1_EF in lieu

of UG25ABV, because working while in college probably correlates with a worse educational experience, and there could be an outside confounder that influence working during undergrad, and influences career earnings.

```
cor(df$UG25ABV, df$PFTFTUG1_EF)
```

```
[1] -0.7890058
```

```
summary(lm(UG25ABV ~ PFTFTUG1_EF, data = df))
```

Call:

```
lm(formula = UG25ABV ~ PFTFTUG1_EF, data = df)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.18309	-0.03475	-0.00579	0.02390	0.36326

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.476228	0.001611	295.7	<2e-16 ***
PFTFTUG1_EF	-0.491355	0.002182	-225.2	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.05803 on 30739 degrees of freedom

Multiple R-squared: 0.6225, Adjusted R-squared: 0.6225

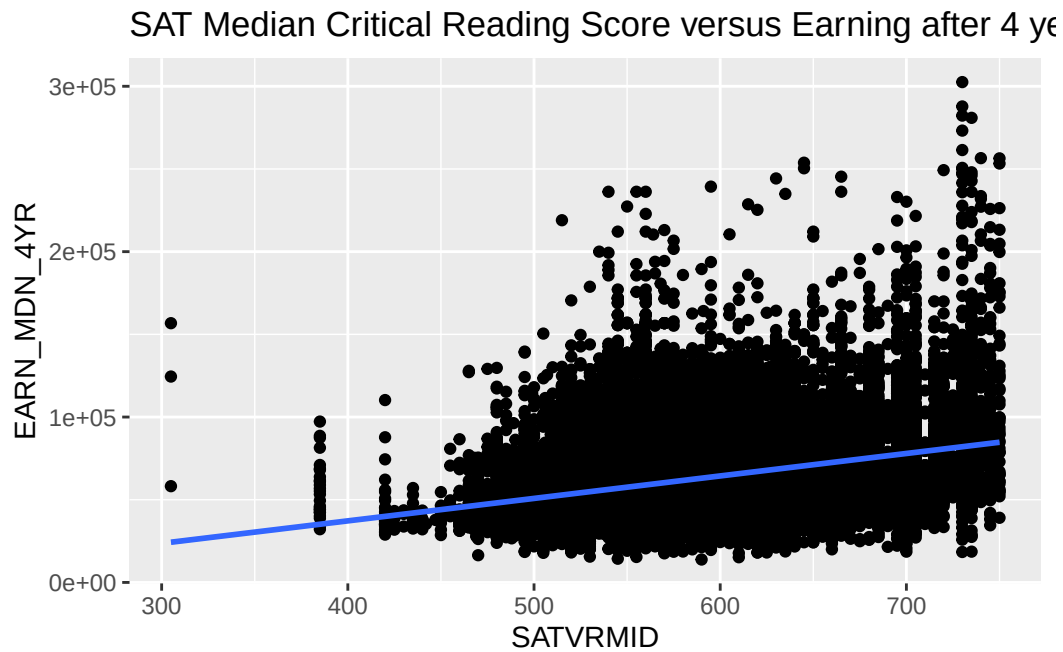
F-statistic: 5.07e+04 on 1 and 30739 DF, p-value: < 2.2e-16

Appendix

SAT Scores

```
ggplot(df, aes(x=SATVRMID, y = EARN_MDN_4YR)) +  
  geom_point()+  
  labs(  
    title = "SAT Median Critical Reading Score versus Earning after 4 years"  
  ) +  
  geom_smooth(method="lm")
```

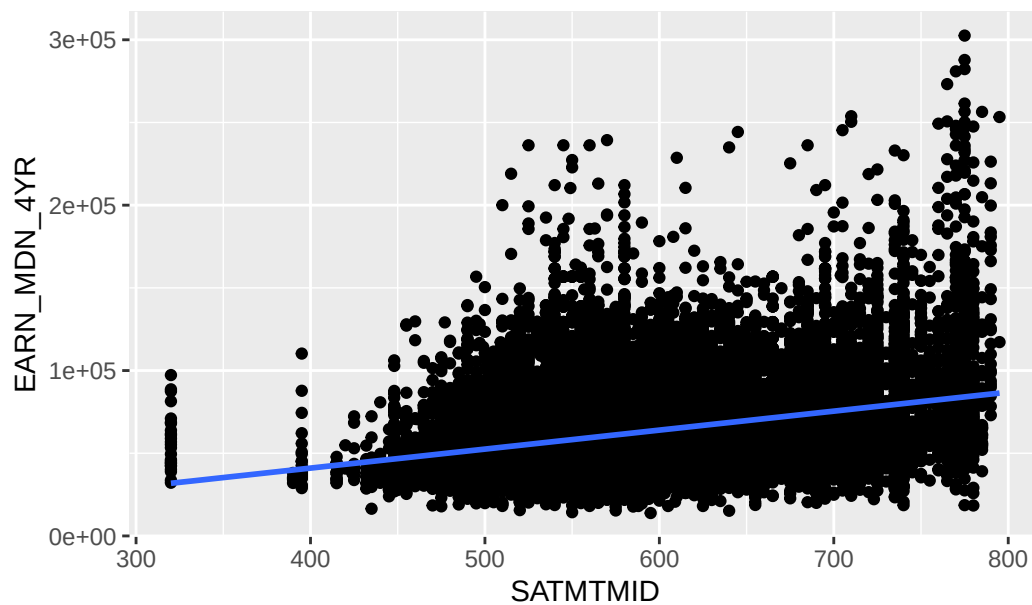
```
`geom_smooth()` using formula = 'y ~ x'
```



```
ggplot(df, aes(x=SATMTMID, y = EARN_MDN_4YR)) +  
  geom_point()+  
  labs(  
    title = "SAT Median Math Score versus Earning after 4 years"  
  ) +  
  geom_smooth(method="lm")
```

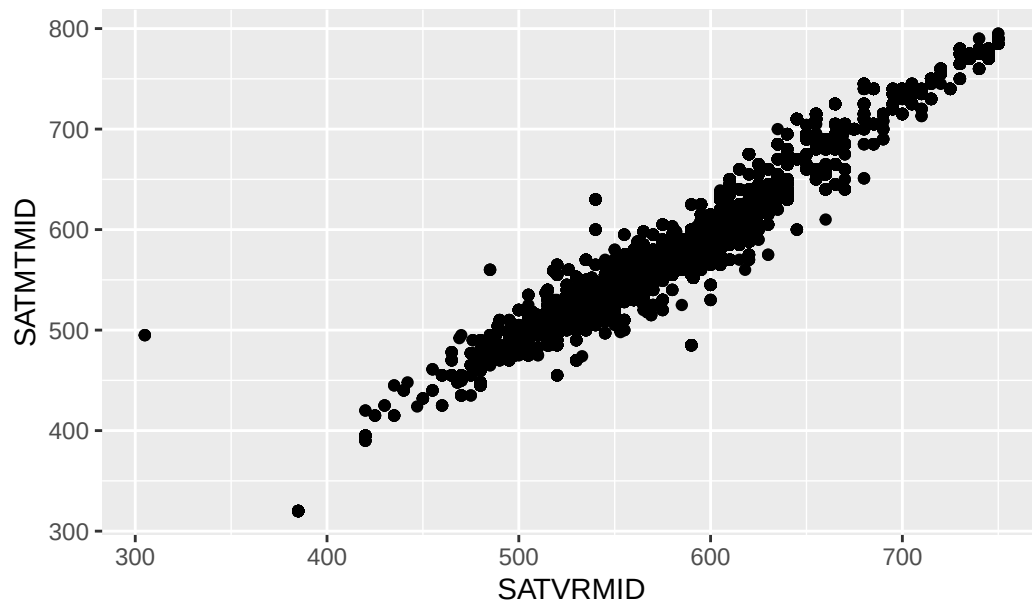
```
`geom_smooth()` using formula = 'y ~ x'
```

SAT Median Math Score versus Earning after 4 years



```
ggplot(df, aes(x=SATVRMID, y = SATMTMID)) +  
  geom_point()+  
  labs(  
    title = "SAT Median Critical Reading Score versus SAT Median Math Score"  
  )
```

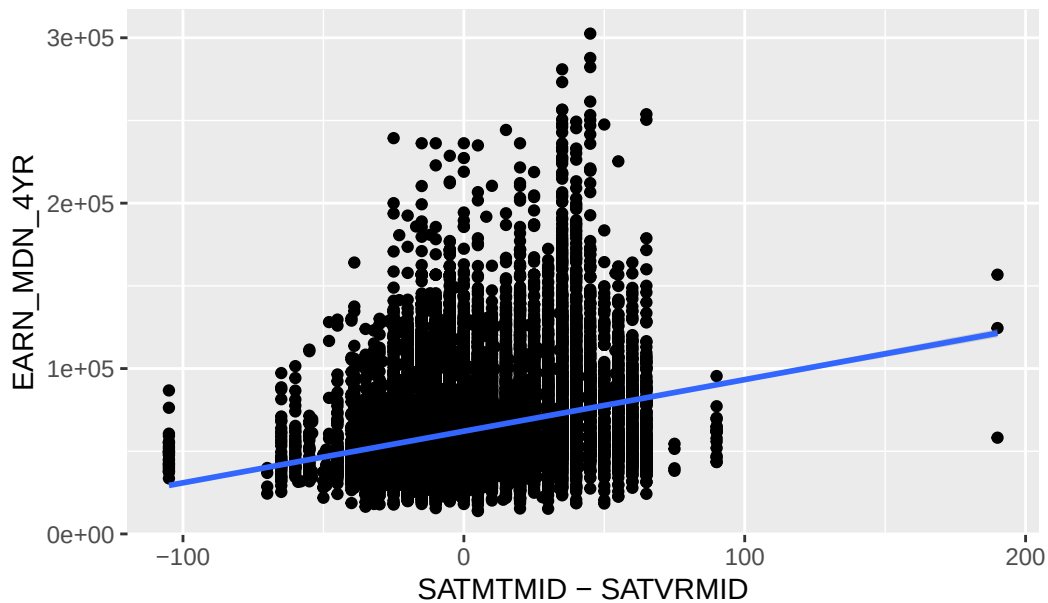
SAT Median Critical Reading Score versus SAT Median Math Score



```
ggplot(df, aes(x=SATMTMID - SATVRMID, y = EARN_MDN_4YR)) +
  geom_point()+
  labs(
    title = "SAT Median Math Minus SAT Median Critical Thinking versus Earnings"
  ) +
  geom_smooth(method="lm")
```

`geom\_smooth()` using formula = 'y ~ x'

SAT Median Math Minus SAT Median Critical Thinking versu:



```
simple_df <- tibble(
  Y=df$EARN_MDN_4YR,
  X1=df$SATMTMID + df$SATVRMID,
  X2=df$SATMTMID - df$SATVRMID
)
sat_model <- lm(Y~X1+X2, simple_df)
summary(sat_model)
```

Call:

```
lm(formula = Y ~ X1 + X2, data = simple_df)
```

Residuals:

Min	1Q	Median	3Q	Max
-66524	-15927	-5632	11032	218386

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2606.685	1761.682	1.48	0.139
X1	50.954	1.504	33.88	<2e-16 ***
X2	107.734	8.916	12.08	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

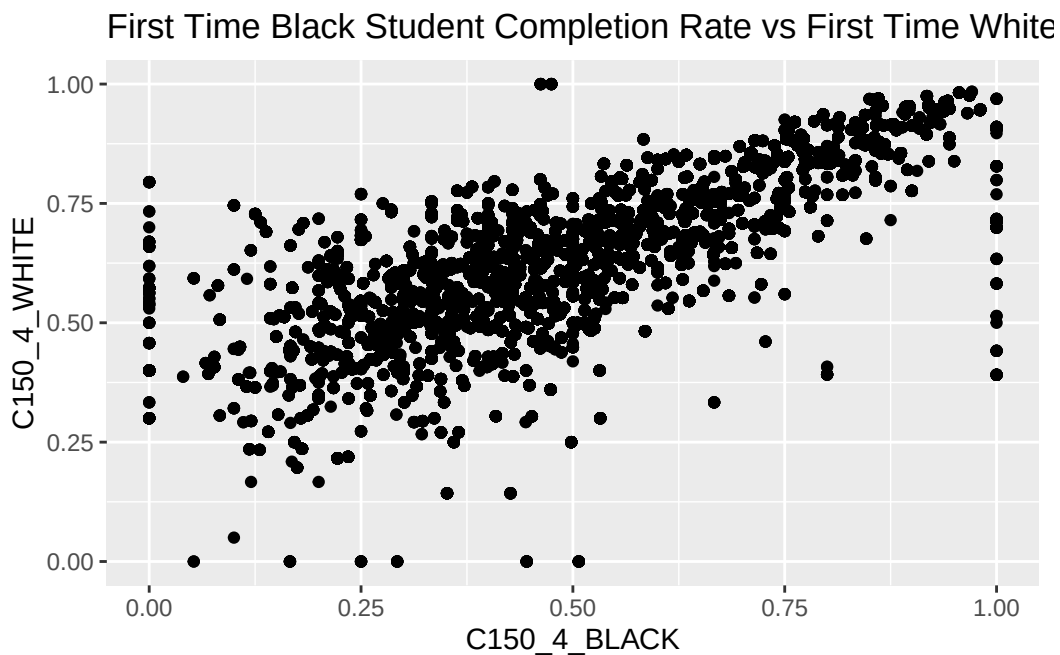
Residual standard error: 24170 on 30738 degrees of freedom

Multiple R-squared: 0.09969, Adjusted R-squared: 0.09964

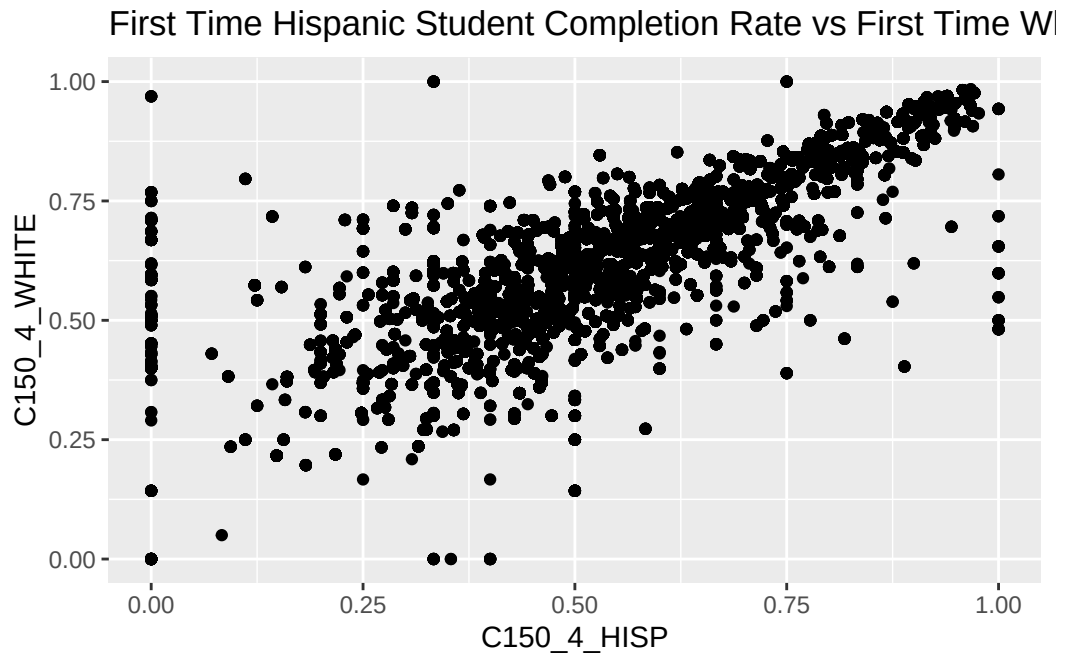
F-statistic: 1702 on 2 and 30738 DF, p-value: < 2.2e-16

First Time Completion Rates By Demographic

```
ggplot(df, aes(x=C150_4_BLACK , y = C150_4_WHITE)) +  
  geom_point()+  
  labs(  
    title = "First Time Black Student Completion Rate vs First Time White Student Completion  
  )
```



```
ggplot(df, aes(x=C150_4_HISP , y = C150_4_WHITE)) +  
  geom_point()+  
  labs(  
    title = "First Time Hispanic Student Completion Rate vs First Time White Student Complet.  
  )
```



```
ggplot(df, aes(x=C150_4_HISP , y = C150_4_BLACK)) +  
  geom_point()+  
  labs(  
    title = "First Time Hispanic Student Completion Rate vs First Time Black Student Complet.  
  )
```

